

Problem A. Koshary

Time Limit 1000 ms

Mem Limit 262144 kB

Yousef is at the coordinates $(0, 0)$ and wants to reach a plate of Koshary at (x, y) .

To get there, Yousef takes long steps. From any point (a, b) , a long step moves him to:

- $(a + 2, b)$ or $(a, b + 2)$

However, Yousef is allowed to take **at most one** short step during his entire journey. A short step moves him to:

- $(a + 1, b)$ or $(a, b + 1)$

Can Yousef reach the exact coordinates (x, y) of the Koshary plate?

Input

The first line contains an integer t ($1 \leq t \leq 100$) — the number of test cases.

Each test case contains two integers x and y ($1 \leq x, y \leq 10$) — the coordinates of the Koshary plate.

Output

For each test case, output "YES" if Yousef can reach the Koshary plate and "NO" otherwise.

You can output the answer in any case (upper or lower). For example, the strings "yEs", "yes", "Yes", and "YES" will be recognized as positive responses.

Examples

Input	Output
6 1 1 1 2 4 6 5 9 7 2 10 10	NO YES YES NO YES YES YES

Note

In the first test case, it is impossible to reach (1, 1) using the mentioned steps.

In the second test case, an optimal sequence of steps would be:

$$(0, 0) \xrightarrow{\text{long}} (0, 2) \xrightarrow{\text{short}} (1, 2)$$

In the third test case, an optimal sequence of steps would be:

$$(0, 0) \xrightarrow{\text{long}} (0, 2) \xrightarrow{\text{long}} (2, 2) \xrightarrow{\text{long}} (2, 4) \xrightarrow{\text{long}} (4, 4) \xrightarrow{\text{long}} (4, 6)$$